SAMRAT ASHOK TECHNOLOGICAL INSTITUTE

(Engineering College),VIDISHA (M.P.)

(An Autonomous Institute Affiliated to RGPV Bhopal)

Department of Information Technology

Semester/Year			Program		B.Tech.
Subject Category	DLC	Subject Code:	IT26103P	Subject Name:	Object Oriented Programming Technology

Maximum Marks Allotted						Contact Hours			Total Credits
Theory			Practical		Total Marks				
End Sem	Mid - Sem	Assignment/ Quiz	End Sem	Lab- Work		L	T	P	
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Pre-requisites:

1. Basic knowledge of computer programming concepts.
2. Understanding of problem-solving techniques and algorithms.
3. Familiarity with procedural programming using languages such as C.
4. Knowledge of data types, operators, control structures, and functions.
5. Basic understanding of compilation and execution process.

Course Objective:

1. Introduce the fundamental concepts and principles of Object-Oriented Programming using C++.
2. Develop the ability to design and implement programs using classes, objects, inheritance, and polymorphism.
3. Enable students to apply constructors, destructors, operator overloading, and friend functions in software development.
4. Provide knowledge of templates, exception handling, and runtime type identification techniques.
5. Familiarize students with file handling, streams, and the Standard Template Library (STL) for efficient programming practices.Enhance problem-solving and software design skills using object-oriented methodologies.

Course Outcomes:

Upon completion of this course, the student will be able to:

CO1: Explain the core concepts of Object-Oriented Programming including abstraction, encapsulation, inheritance, and polymorphism.

CO2: Develop programs using classes, objects, constructors, destructors, and operator overloading techniques.

CO3: Apply templates and exception handling mechanisms to build reusable and reliable software solutions.

CO4: Implement runtime polymorphism, virtual functions, RTTI, and advanced inheritance concepts in applications.

CO5:Use streams, file handling, and STL components effectively for developing efficient C++ programs. Design and debug object-oriented applications using appropriate programming practices and coding standards.

UNITs	Descriptions	CO's
I	**Introduction:** Object oriented programming concepts – Objects – Classes – Abstraction andencapsulation – inheritance – Abstract classes – Polymorphism– Function overloading, Operatoroverloading – Friend functions – Overloading through friend functions - Constant and volatilefunctions.	CO1
II	**Constructors and Destructors:** Constructors – Default constructor – Parameterizedconstructors – Constructor with dynamic allocation – Copy constructor – Destructors– Typeconversion – Explicit constructor.	CO2
III	**Templates and Exceptions:** Function and Class templates - Exception handling.	CO3
IV	**Inheritance and RTTI:** Inheritance - virtual base class – abstract class – Runtimepolymorphism – virtual functions – pure virtual functions – RTTI – Dynamic casting – RTTI andtemplates – cross casting – down casting	CO4
V	**Streams, I/O and Files:** Streams and Formatted I/O – I/O manipulators - File handling – ANSIString Objects –Standard Template Library (STL).	CO5
Guest Lectures(if any)		--
Total Hours		

Text Books: -

1. Herbert Schildt, C++ Complete Reference, 4th Edition, McGraw Hill, 2002.
2. HM Deitel and PJ Deitel, C++ How to Program, 7th Edition, Prentice Hall, 2010.
3. Object oriented programming with C++ by Balaguruswamy. TMH Publishing

REFERENCESBooks: -

1. Ira Pohl, Object Oriented Programming using C++, 2nd Edition, Pearson Education,2004.
2. S. B. Lippman, Josee Lajoie, Barbara E. Moo, C++ Primer, 4th Edition, PearsonEducation, 2005.
3. B. Stroustrup, The C++ Programming Language, 3rd edition, Pearson Education, 2004.
4. C++ Language Tutorial: http://www.cplusplus.com/doc/tutorial.
5. Wikiversity: Introduction to C++ :https://en.wikiversity.org/wiki/C%2B%2B/Introduction
6. B. Trivedi, Programming with ANSI C++, Oxford University Press, 2007.
7. Learn about C++ Programming: http://cplus.about.com/od/learning1.

Modes of Evaluation and Rubric:

The evaluation modes consist of performance in lab work and end-semester practical examinations.

List/Links of e-learning resource:

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2
CO-1	3	2	1		2					1		2	3	2
CO-2	3	3	3	1	3				1	1		2	3	3
CO-3	3	3	3	2	3			1	1	1		2	3	3
CO-4	3	3	3	2	3				1	1		2	3	3
CO-5	3	3	3	2	3			1	1	2	1	2	3	3

List of Experiments:

1. a. Develop a C++ program using classes and member functions to accept a paragraph, print each word in the paragraph with the number of vowels in each word.
   b. A company produces pens. Three salesmen of the company are selling the pens in four different districts. Develop a C++ program using classes and member functions to read the quantity sold by each salesman in different districts and display the quantity sold by each sales man with district and the total quantity sold.
   c. Develop a C++ program using classes and member functions to perform the arithmetic operations on matrices.
2. Design C++ classes with static members, methods with default arguments, friend functions. (For example, design matrix and vector classes with static allocation, and a friend function to do matrix-vector multiplication)
3. Implement complex number class with necessary operator overloading and type conversions such as integer to complex, double to complex, complex to double etc.
4. Write a C++ program to perform the string concatenation using dynamic memory allocation.
5. Implement Matrix class with dynamic memory allocation and necessary methods. Give proper constructor, destructor, copy constructor, and overloading of Assignment operator overloads the new and deletes operators to provide custom dynamic allocation of memory.
6. Develop templates of standard sorting algorithms such as bubble sort, insertion sort.
7. Exception handling - Divide by Zero, arrays out of bounds, memory exhaustion exception:
   a. Insertion and selection sort on (i) integer array (ii) strings
   b. Linear Search and Binary search over (i) integer array (ii) strings
8. a. Define Point class and an Arc class. Define a Graph class which represents graph as a collection of Point objects and Arc objects. Write a method to find a minimum cost spanning tree in a graph.
   b. Write a C++ program with two classes named as "one", "two" to find the area of square and rectangle respectively, and inherit these values to another class named as "cuboid" with its own property "height" to find the area of the cuboid.
9. Develop with suitable hierarchy, classes for Point, Shape, Rectangle, Square, Circle, Ellipse, Triangle, Polygon, etc. Design a simple test application to demonstrate dynamic Polymorphism and RTTI.
10. Write a C++ program that randomly generates complex numbers (use previously designed Complex class) and writes them two per line in a file along with an operator (+, -, *, or /). The numbers are written to file in the format (a + ib). Write another program to read one line at a time from this file, perform the corresponding operation on the two complex numbers read, and write the result to another file (one per line).
11. a. Develop a simple C++ vector container program.
    b. Implement C++ STL vector using various operators code.
    c. Implement C++ STL vector using constructors

Recommendation by Board of studies on	
Approval by Academic council on	
Designed by	IT Department